

European Championship in Trustworthy AI

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Title: FL Data Reconstruction from Gradients

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Task: Reconstruct training images from provided gradients in a federated learning setting

Federated learning (FL) enables collaborative model training without sharing raw data, instead relying on merely sharing gradients generated by training the provided models on local data. However, while this procedure seems secure, it has been shown that training data can be reconstructed from the gradients alone. This makes it possible for an honest-but-curious server to gain access to the clients' private training data. The effect can even be enhanced if the server maliciously applies so-called "trap weights" to the provided model.

In this task, you are in the shoes of such a malicious server. Your goal is to reconstruct training images as accurately as possible, across a number of models and gradients with different properties and defenses.

As the server, you get:

- Full white-box access to the global models that should be trained in the FL process, and that were used to generate the gradients
- The gradients a client generated on each of the models

All data is available on [HuggingFace](#).

[hackathon_setup.sh](#) provided to you creates your team folder, downloads the datasets and creates a per task environment for you.

What You Don't Have Access To

- The private data used by the client, or their corresponding labels.

What's your task:

There are 12 models, and each model was used by one client to generate gradients. This client in turn used 128 private images to calculate the gradient. This yields a total of 1536 used private images. It is your goal to reconstruct these 1536 images as accurately as possible.

Your Starting Point

To start you off, we give you a [task template](#) that shows you how to access the models and gradients, as well as instructions explaining the result submission process.

How to submit your results?

- We will provide a unique API-key to every participating team. You are supposed to only use this token to submit your results to our server. If you did not receive this team-specific token, please contact us.
- Your submissions should consist of a .pt file containing exactly 1536 reconstructed images. Exactly 128 images should be provided per model in any order. The file should contain a dictionary, structured as follows:

```
{  
  "model1":  
    Tensor( 128 images )  
  "model2":  
    Tensor (128 images )  
  ...  
}
```

- Submitted images **MUST** have a resolution of 64 x 64 pixels, have dtype float32, and all pixel values have to be in range [0,1]. Per model, the reconstructions should be provided as a Tensor of shape (128, 3, 64, 64).
- You are supposed to submit your images using the code provided in the coding template. Remember to replace [YOUR_API_KEY_HERE](#) with your actual API-key (keep the double quotes), and replace [FILE_PATH](#) with your real results. Also set [SUBMIT](#) to True.
- To avoid the possibility of brute-forcing, only one submission can be made every 5 minutes. If the submission was unsuccessful due to an error, this cooldown period will only be 2 minutes.

Evaluation Metric

After submitting your results, they will be evaluated. The evaluation metric pairs each submitted image up with the training image from the corresponding model that is closest to it in terms of structural similarity index (SSIM). The average SSIM across **all** images will be converted into range [0,1]. This is the final score.

Note - No two submitted images are paired up with the same training image, so duplicating well reconstructed images in the submission does not achieve optimal results.

Leaderboard

After evaluation, your results can be found in the leaderboard:

- You can access the leaderboard for this task at

http://35.192.205.84/leaderboard_page. This will help you to compare your solutions with other teams and see where you stand.

- The leaderboard shows the best result per team only. If your current submission's score is lower than the score saved in the leaderboard, the score will not be updated.
- The leaderboard only shows you the evaluation on 30% of the data. After the hackathon is concluded, you will be ranked based on the other 70%. This is done to avoid the possibility of overfitting, and promote methods that properly generalize.

References

- "When the curious abandon honesty: Federated learning is not private" Franziska Boenisch, Adam Dziedzic, Roei Schuster, Ali Shahin Shamsabadi, Ilia Shumailov, Nicolas Papernot. Euro S&P 2023. <https://arxiv.org/pdf/2112.02918> (Analytic Reconstruction, trap weights)
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- "Inverting Gradients - How easy is it to break privacy in federated learning?" Jonas Geiping, Hartmut Bauermeister, Hannah Dröge, Micheal Moeller. NeurIPS 2020. <https://arxiv.org/pdf/2003.14053> (Reconstruction via Optimization)